

# Skills Summary

## Taylor Polynomials and Remainder Bounds

Use this reference while completing your worksheet or when studying.

### Skill 1 — Building the polynomial from derivatives

**Rule:**  $P_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k$ . Each coefficient needs one derivative evaluated *at the centre*, divided by that order's factorial. Work down the list of derivatives once and record the values at  $a$ ; the powers of  $(x-a)$  take care of themselves.

**Example:** For  $f(x) = e^{3x}$ , find  $f'''(x)$  and hence the cubic coefficient of the Maclaurin polynomial.

**Step 1.** The cubic coefficient is the third derivative at zero divided by three factorial, so the only real work is differentiating three times.

**Step 2.** Each differentiation multiplies by 3, so  $f'''(x) = 27e^{3x}$ , giving  $f'''(0)/3! = 27/6$ .  
 $\Rightarrow 27e^{3x}$

**Try it:** For  $f(x) = e^{4x}$ , find  $f'''(x)$ .

check:  $64e^{4x}$

### Skill 2 — Reusing a series you already know

**Four to memorise:**  $e^x = \sum \frac{x^n}{n!}$ ,  $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$ ,  $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$ ,  $\frac{1}{1-x} = 1 + x + x^2 + \dots$ . To get a new series, *substitute* into one of these, or multiply it by a power of  $x$  — never differentiate from scratch when a substitution will do.

**Example:** Estimate  $\sin(0.3)$  using terms through  $x^5$ .

**Step 1.** The sine series is already known, so substitute the input rather than building anything — three terms is plenty at this size.

**Step 2.**  $0.3 - \frac{0.3^3}{6} + \frac{0.3^5}{120} = 0.3 - 0.0045 + 0.00002$ .  
 $\Rightarrow 0.29552$

**Try it:** Estimate  $\sin(0.2)$  using terms through  $x^5$ .

check:  $0.19866$

### Skill 3 — Bounding the error

**Two bounds, and the condition that chooses between them.** Lagrange:  $|R_n(x)| \leq \frac{M |x-a|^{n+1}}{(n+1)!}$ , where  $M$  bounds  $|f^{(n+1)}|$  across the whole interval — always available. Alternating: the error is at most the first omitted term — available only when the series alternates and its term sizes decrease to zero. Pick  $M$  by asking where the next derivative is largest on the interval.

**Example:** Bound the error when  $P_2$  of  $e^x$  estimates  $e^x$  at  $x = 1$ .

**Step 1.** The series for the exponential does not alternate at a positive input, so the Lagrange form is the one available, and it needs a bound on the third derivative.

**Step 2.** Every derivative is  $e^x$ , largest at  $x = 1$ , so  $M = e$  and the bound is  $\frac{e \cdot 1^3}{3!}$ .

$$\Rightarrow \frac{e}{6} \approx 0.45305$$

**Try it:** Bound the error when  $P_4$  of  $e^x$  estimates  $e^x$  at  $x = 1$ .

check:  $\frac{120}{e} \approx 0.02265$